

CY 202P: Organic Chemistry Laboratory

Course Code	CY 202P
Course Name	Organic Chemistry Laboratory
L-T-P-C	0-0-4-2
Prerequisites	None
Intended for	BS Chemical Sciences
Mutual Exclusion	None
Approval	42nd Senate (Jan 2024)

Course Contents

- **Module I: Basic Lab Techniques**
 1. To determine type and detection of extra elements (N, S, Cl, Br, I) in organic compounds
 2. Thin layer chromatography (TLC) and calculation of R_f values
 3. Separation of 2 organic compounds by paper chromatography
 4. Purification of organic compounds by crystallization.
- **Module II: Organic Qualitative Analysis**
 1. Separation of two components from the binary mixture of organic compounds (Solid-Solid, Solid-Liquid).
- **Module III: Organic Preparations** (any 6)
 1. Preparation of paracetamol
 2. Preparation of aspirin
 3. Preparation of phthalimide from phthalic anhydride
 4. Preparation of 2:4-DNP derivative of aldehyde or ketone
 5. Preparation 4-chloro benzyl alcohol from 4-chloro benzaldehyde
 6. Base catalyzed Aldol condensation
 7. Preparation 4-iodonitrobenzene from 4-nitroaniline by Sandmeyer Reaction
 8. Preparation of Glucosazone derivative of Glucose
 9. Preparation of quinone from hydroquinone
 10. Preparation of Oxime derivative of Ketones.

Text Books

1. **Vogel's book of Practical Organic Chemistry**, 5th Edition, Longman Scientific & Technical, 2006.
2. Pavia, Lampman, Kriz & Engel, **Organic Chemistry A Lab Manual**, Cengage Learning, 2009.

References

1. Frederick A. Bettelheim, **Experimental physical chemistry**, Saunders, 1971.
2. A. Ghosal, B. Mahapatra, A. K. Nad, **An Advanced Course in Practical Chemistry**, New Central Book Agency Pvt Ltd, 2000.